

Anexo N — Lean: invariantes de la cronología

Generado desde el repositorio storyMaker · commit 014fb99 · 2026-09-25 18:45 UTC · Qaracter group 2026

Source: docs/spec.md §1 (Lean 4 chronology validator), §2 (validator lean_chronology, kind c), §5 (Lean runs only if elan installs without administrator rights) and AC-11.

What is here

Chronology.lean states the example novel's timeline (run 02412b7fe29e, the-other-side-of-the-hill) as concrete Lean lists and proves two invariants by decide:

theorem	what it says
events_in_order	the thirteen events of the timeline are in chronological order (day, then moment of the day)
story_within_one_week	the whole story happens within one week, so no stated age changes between events
ages_possible	every stated age (el destinatario 10, Rosa 10, Elena 12) is between 0 and 120

The events and ages are transcribed from the run's own Story Bible, output/the-other-side-of-the-hill/bible/timeline.md and characters.md.

Status: not run

- lake build / lean Chronology.lean has not been run. elan, Lean's installer, needs administrator rights on the build machine, which the project does not have. This is the reason the spec admits for AC-11, and it is recorded as lean_chronology: not run: elan unavailable in validations.
- The file was transcribed by hand, not generated. The exporter the spec names (backend/formal/lean_export.py, from the SQLite chronology table) was not written: it was the second cut in PLAN-001 Part 6. The file therefore proves the invariants over this one run's chronology, not over every run.
- What it would not catch today. Brief 05's impossible dates are refused earlier, at FLOW-0 (a memory dated before the birth date). A wedding at nine is a possible date but an implausible age; catching it would need an invariant on the age at a named event, which this file does not state.

To run it where elan is available

```
lean Chronology.lean
```

No output and exit code 0 means the three theorems hold.